

Algebra 1

Unit 1

Lesson 10 Homework

Name_____

Date_____

Period_____

1. To multiply two radicals with the same index, multiply the _____ together and then multiply the _____ together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{13} \cdot \sqrt{13}$

3. $\frac{\sqrt{5}}{4\sqrt{125}}$

4. $\sqrt[3]{6} \cdot \sqrt[3]{9}$

5. $\sqrt{24} \cdot \sqrt{8}$

6. $\frac{\sqrt[3]{375}}{\sqrt[3]{648}}$

7. $\sqrt[3]{18} \cdot \sqrt[3]{60}$

8. Identify all equivalent expressions to $\sqrt{15} \cdot \sqrt{6}$.

- ☐ $9\sqrt{10}$
- ☐ $\sqrt{90}$
- ☐ $(10 \cdot 9)^{\frac{1}{2}}$
- ☐ $3\sqrt{15}$
- ☐ $3\sqrt{10}$
- ☐ $15^{\frac{1}{3}} \cdot 6^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{2\sqrt{6}}{\sqrt{5}}$

10. $\frac{\sqrt{28}}{3\sqrt{8}}$